2O13

SHAGUN GARG

1024150104

Lab Assignment 4

Constructor, destructor, dynamic memory management

1. Create a class named ***'Rectangle'*** with two data members- *length* and *breadth* and a function to calculate the area which is 'length*breadth'. The class has *three constructors* which are :

   (a) having no parameter - values of both length and breadth are assigned zero.

   (b) having two numbers as parameters - the two numbers are assigned as length and breadth respectively.

   (c) having one number as parameter - both length and breadth are assigned that number.

   Now, create objects of the 'Rectangle' class having none, one and two parameters and print their areas.

```
Users > Shagun Garg > oops.cpp > rectan
#include <iostream>
#include <string>
using namespace std;
class rectangle{
    int length;
    int bredth;

    public:
    rectangle(){
        length=0;
        bredth=0;
    }
    rectangle(int l,int b){
        length=l;
        bredth=b;
    }
    rectangle(int n){
        length=n;
        bredth=n;
    }
    void area(){
        int a=length*bredth;
        cout<<a;
        cout<<endl;
    }
};
```

```
int main(){
    rectangle r1;
    rectangle r2(6);
    rectangle r3(4,7);
    cout<<"Area of rectangle(0 Parameter)= ";
    r1.area();
    cout<<"Area of rectangle(1 Parameter)= ";
    r2.area();
    cout<<"Area of rectangle(2 Parameter)= ";
    r3.area();
}
```

```
PS C:\Users\Shagun Garg> cd "c:\Users\Shagun Garg
Area of rectangle(0 Parameter)= 0
Area of rectangle(1 Parameter)= 36
Area of rectangle(2 Parameter)= 28
PS C:\Users\Shagun Garg>
```

2. Redefine the above program by creating an array of objects of the class Rectangle and calculate area for each object calling different constructors. Also implement constructors with default arguments and destructor in this program.

```
Users > Shagun Garg > oops.cpp > rectangle > ~re
#include <iostream>
#include <string>
using namespace std;
class rectangle{
    int length;
    int bredth;

    public:
    rectangle(){
        length=0;
        bredth=0;
    }
    rectangle(int l,int b){
        length=l;
        bredth=b;
    }
    rectangle(int n){
        length=n;
        bredth=n;
    }
    void area(){
        int a=length*bredth;
        cout<<a;
        cout<<endl;
    }
    ~rectangle(){
        cout<<"Destructor called."<<endl;
    }
};
```

```
int main(){
    rectangle r[3]={rectangle(),rectangle(3),rectangle(6,2)};
    cout<<"Area of rectangle(0 Parameter)= ";
    r[0].area();
    cout<<"Area of rectangle(1 Parameter)= ";
    r[1].area();
    cout<<"Area of rectangle(2 Parameter)= ";
    r[2].area();
}
```

```
PS C:\Users\Shagun Garg> cd "c:\Users\Shagun Garg\"
Area of rectangle(0 Parameter)= 0
Area of rectangle(1 Parameter)= 9
Area of rectangle(2 Parameter)= 12
Destructor called.
Destructor called.
Destructor called.
PS C:\Users\Shagun Garg>
```

3. Verify the following about *destructor* by writing the program:

(i) Name should begin with tilde sign(~) and must match class name.

```
Users > Shagun Garg > oops.cpp > rectangle > ~re
#include <iostream>
#include <string>
using namespace std;
class rectangle{
    int length;
    int bredth;

    public:
    rectangle(){
        length=0;
        bredth=0;
    }
    rectangle(int l,int b){
        length=l;
        bredth=b;
    }
    rectangle(int n){
        length=n;
        bredth=n;
    }
    void area(){
        int a=length*bredth;
        cout<<a;
        cout<<endl;
    }
    ~rectangle(){
        cout<<"Destructor called."<<endl;
    }
};
```

```
    ~rectangle(){
        cout<<length<<","<<bredth<<endl;
        cout<<"Destructor called."<<endl;
    }

};
int main(){
    rectangle r[3]={rectangle(),rectangle(3),rectangle(6,2)};
    cout<<"Area of rectangle(0 Parameter)= ";
    r[0].area();
    cout<<"Area of rectangle(1 Parameter)= ";
    r[1].area();
    cout<<"Area of rectangle(2 Parameter)= ";
    r[2].area();
}
```

```
PS C:\Users\Shagun Garg> cd "c:\Users\Shagun Garg\"
Area of rectangle(0 Parameter)= 0
Area of rectangle(1 Parameter)= 9
Area of rectangle(2 Parameter)= 12
6,2
Destructor called.
3,3
Destructor called.
0,0
Destructor called.
PS C:\Users\Shagun Garg>
```

(ii) There cannot be more than one destructor in a class.

```
    }
    void area(){
        int a=length*bredth;
        cout<<a;
        cout<<endl;
    }
    ~rectangle(){
        cout<<"Destructor called."<<endl;
    }
    ~rectangle(){
        cout<<"Second destructor."<<endl;
    }
};
```

By using 2 destructors in a class,

Error occurred due to overloading of destructors.

OUTPUT:

```
PS C:\Users\Shagun Garg> cd "c:\Users\Shagun Garg\" ; if ($?) { g++ oops.cpp -o oops } ; if ($?) { .\oops }
oops.cpp:29:5: error: 'rectangle::~rectangle()' cannot be overloaded
    ~rectangle(){
    ^
oops.cpp:26:5: error: with 'rectangle::~rectangle()'
    ~rectangle(){
    ^
PS C:\Users\Shagun Garg>
```

(iii) Destructors do not allow any parameter.

INPUT:

```
~rectangle(int l,int b){
    length=l;
    bredth=b;
    cout<<l<<" "<<b<<endl;
    cout<<"Destructor called."<<endl;
}
```

OUTPUT:

```
oops.cpp:26:27: error: destructors may not have parameters
     ~rectangle(int l,int b){
                            ^
oops.cpp: In destructor 'rectangle::~rectangle()':
oops.cpp:27:16: error: 'l' was not declared in this scope
         length=l;
                ^
oops.cpp:28:16: error: 'b' was not declared in this scope
         bredth=b;
                ^
PS C:\Users\Shagun Garg>
```

(iv) They do not have any return type, just like constructors.

When you do not specify any destructor in a class, compiler generates a default destructor and inserts it into your code.

INPUT:

```
~rectangle(){
    cout<<length<<","<<bredth<<endl;
    cout<<"Destructor called."<<endl;
    return length;
}
```

OUTPUT:

```
PS C:\Users\Shagun Garg> cd "c:\Users\Shagun Garg\" ; if ($?) {
oops.cpp: In destructor 'rectangle::~rectangle()':
oops.cpp:29:16: error: returning a value from a destructor
         return length;
                ^~~~~~
PS C:\Users\Shagun Garg>
```

4. Implement dynamic memory allocation. Use *new* and *delete* keywords. (For integer variable, float variable, integer array, float array, class objects, Array of Objects)

Users > Shagun Garg > Player.cpp > Box

```cpp
#include <iostream>
using namespace std;

class Box {
public:
    int length;
    void set(int l) {
        length = l;
    }
    void show() {
        cout << "Length = " << length << endl;
    }
};

int main() {
    int* a = new int;
    *a = 10;
    cout << "Integer: " << *a << endl;
    delete a;

    float* b = new float;
    *b = 5.5;
    cout << "Float: " << *b << endl;
    delete b;

    int* arr = new int[3];
    arr[0] = 1;
    arr[1] = 2;
    arr[2] = 3;

    cout << "Integer Array: ";
    for(int i = 0; i < 3; i++)
        cout << arr[i] << " ";
    cout << endl;
    delete[] arr;

    float* farr = new float[2];
    farr[0] = 1.1;
    farr[1] = 2.2;

    cout << "Float Array: ";
    for(int i = 0; i < 2; i++)
        cout << farr[i] << " ";
    cout << endl;
    delete[] farr;

    Box* b1 = new Box;
    b1->set(7);
    b1->show();
    delete b1;

    Box* b2 = new Box[2];
    b2[0].set(3);
    b2[1].set(6);

    for(int i = 0; i < 2; i++)
        b2[i].show();
    delete[] b2;

    return 0;
}
```

```
PS C:\Users\Shagun Garg> cd "c:\Users\Shagun Garg\"
Integer: 10
Float: 5.5
Integer Array: 1 2 3
Float Array: 1.1 2.2
Length = 7
Length = 3
Length = 6
PS C:\Users\Shagun Garg>
```